BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLYPROGRESSREPORT (15/06/2026 – 21/06/2026)

WEEK 5

PROJECT NAME	RADIO PROPAGATION ANALYSIS AND PATH PREDICTION

PROJECTDESCRIPTION :	Project focuses on simulating antenna beam propagation and predicting communication coverage.
GROUP MEMBER :	GARIMA CHOUDHARY, MANASH PRASAR, ARYAN KUMAR
GROUP ID :	2026G265
GROUP GUIDE :	Dr. ABDUL JHUMMARWALA
GROUP COORDINATOR :	GARIMA CHOUDHARY
COLLEGE GUIDE NAME:	ABHILASHA SAKSENA
COLLEGE GUIDE No. :	81231 06272
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COLLEGE NAME :	GATI SHAKTI VISHWAVIDYALAYA

15/06/026	• Studied antenna radiation pattern datasets and verified Theta-Phi-Gain values for visualization processing. (Garima Choudhary) • Implemented Theta-Phi-Gain data parsing and initiated Cartesian coordinate generation for radiation pattern modeling. (Manash Prasar) • Assisted in organizing radiation pattern data and validating dataset consistency. (Aryan Kumar)
16/06/2026	• Analyzed antenna gain distribution and its effect on propagation coverage representation. (Garima Choudhary) • Developed spherical-to-Cartesian coordinate transformation logic for 3D radiation pattern generation. (Manash Prasar) • Assisted in testing generated coordinate outputs and reviewing visualization results. (Aryan Kumar)
17/06/2026	• Studied radiation pattern visualization techniques used in wireless communication systems. (Garima Choudhary) • Generated structured Theta-Phi grids and mapped gain values to three-dimensional coordinates (Manash Prasar) • Assisted in validating generated grid structures and coordinate indexing. (Aryan Kumar)
18/06/2026	• Participated in evaluating generated radiation pattern outputs and identifying visualization issues. (Garima Choudhary) • Implemented triangular mesh generation by connecting neighboring Theta-Phi grid points to create continuous surfaces. (Manash Prasar) • Assisted in testing mesh connectivity and recording observations from generated outputs. (Aryan Kumar)
19/06/2026	• Studied CesiumJS visualization workflow and analyzed rendering behavior of generated radiation patterns. (Garima Choudhary) • Integrated generated mesh structures into CesiumJS and performed 3D radiation pattern visualization experiments. (Manash Prasar) • Assisted in verifying rendered outputs and identifying scaling-related issues. (Aryan Kumar)
20/06/2026	• Assisted in evaluating different visualization parameters for improving radiation pattern representation. (Garima Choudhary) • Performed debugging, scaling optimization, and coverage visualization analysis for antenna radiation patterns in CesiumJS. (Manash Prasar) • Assisted in testing generated propagation models and documenting observed behavior. (Aryan Kumar)
21/06/2026	Holiday (Sunday)

Next Week Plan	• During the next week, the team will focus on improving the antenna radiation pattern visualization by refining mesh generation and coordinate transformation techniques.

REFERENCE:

https://sandcastle.cesium.com/

https://cesium.com/learn/cesiumjs-learn/

https://www.qsl.net/4nec2/

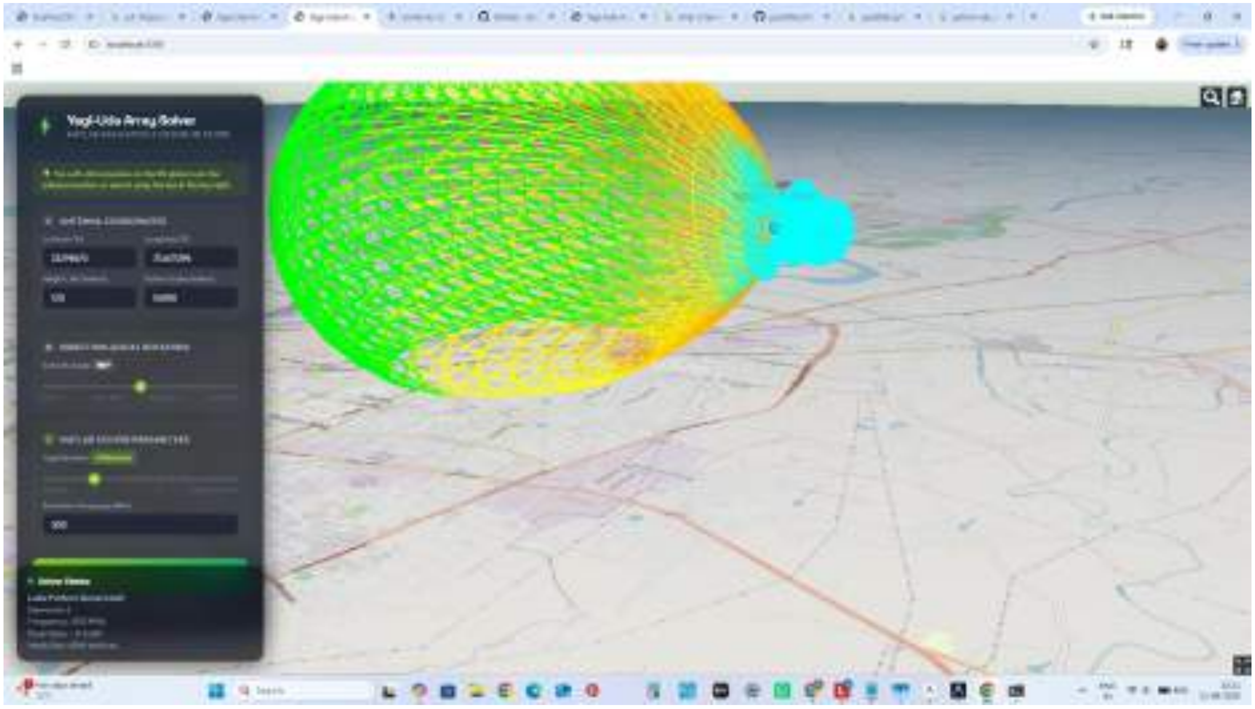
https://www.antenna-theory.com/

https://www.qsl.net/kd2bd/splat.html/

https://www.earthdata.nasa.gov/data/instruments/aster

https://octave.org/doc/interpreter/

SCREEN SHOTS: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